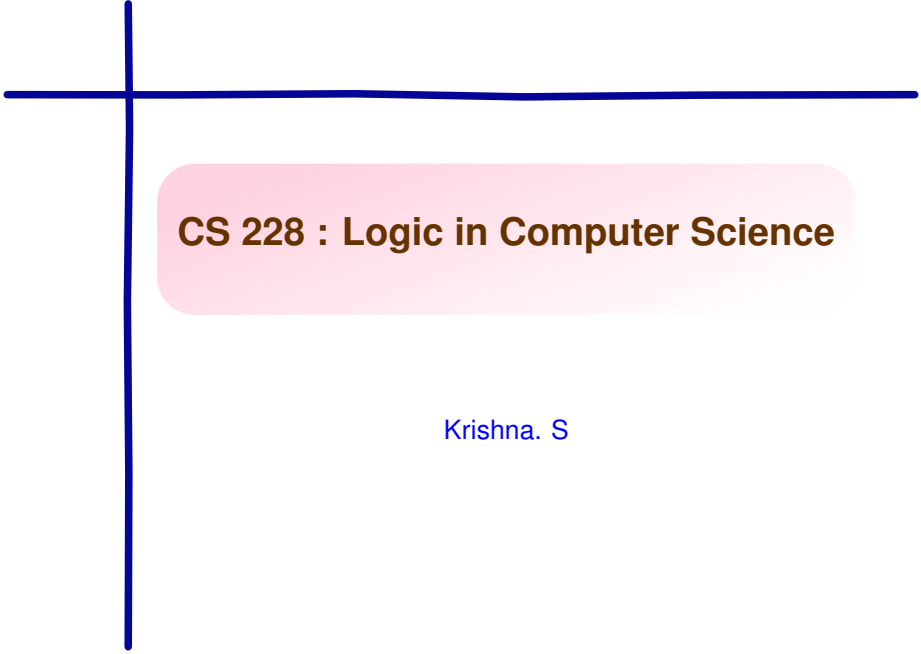




CS 228 : Logic in Computer Science

Krishna. S

Recap

Signatures, Formulae over signatures, terms, and structures to define meanings for the signatures

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- ▶ Recall the signature $\tau = (<, S, Q_a)$ over words. Consider the formula $Q_a(0) \wedge \exists x \exists y (S(x, y) \wedge Q_a(x) \wedge Q_a(y))$. Is this a well formed formula?

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- ▶ Consider the signature τ containing the binary function $+$ and binary relation $<$. Written in infix, $x + 1$ is a term (instead of $+(x, 1)$) and $\forall x (x < (y + 1))$ is a well formed formula.

Recap

We work with first order logic with equality, where, irrespective of the underlying signature, $t_1 = t_2$ is an atomic formula for terms t_1, t_2 .

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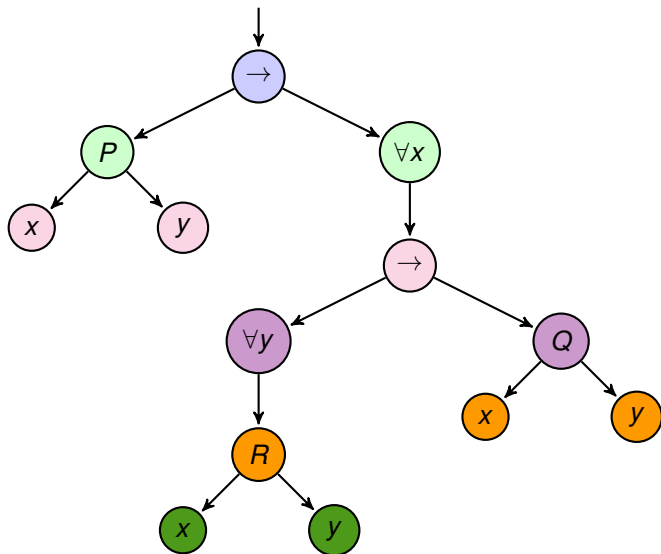
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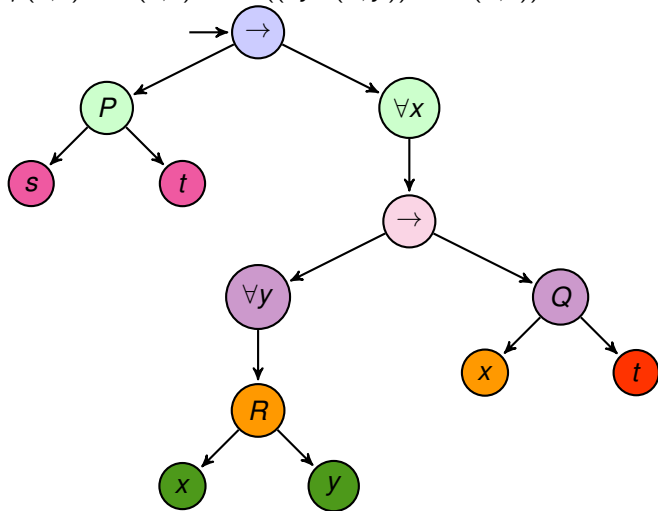
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- ▶ Given φ , denote by $\varphi(x_1, \dots, x_n)$, that $x_1, \dots, x_n$ are the free variables of φ , also $\text{free}(\varphi)$
- ▶ A **sentence** is a formula φ **none** of whose variables are **free**

$$P(x, y) \rightarrow \forall x((\forall y R(x, y)) \rightarrow Q(x, y))$$

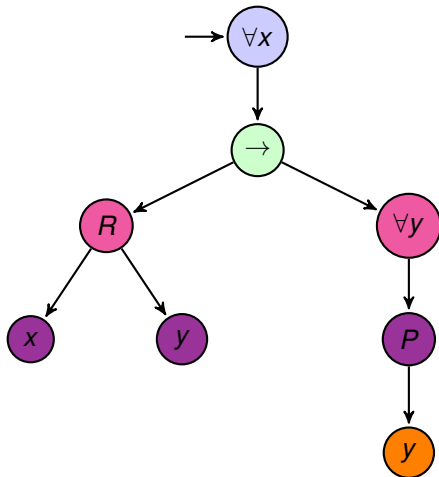


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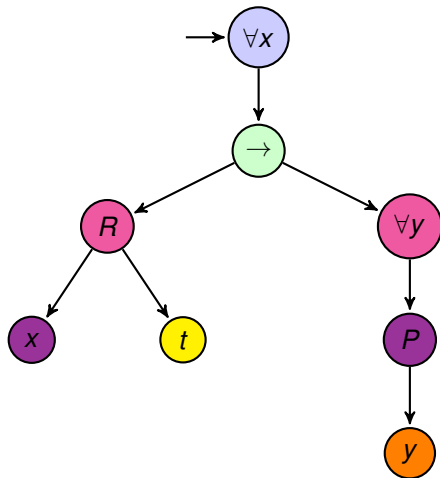
$$\varphi(s, t) = P(s, t) \rightarrow \forall x((\forall y R(x, y)) \rightarrow Q(x, t))$$



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$$\varphi(t) = \forall x(R(x, t) \rightarrow \forall yP(y))$$

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For a τ -structure $\mathcal{A}$, an assignment over $\mathcal{A}$ is a function $\alpha : \mathcal{V} \rightarrow u(\mathcal{A})$ that assigns every variable $x \in \mathcal{V}$ a value $\alpha(x) \in u(\mathcal{A})$. If t is a constant symbol c , then $\alpha(t)$ is $c^{\mathcal{A}}$

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Binding on a Variable

For an assignment α over $\mathcal{A}$, $\alpha[x \mapsto a]$ is the assignment

$$\alpha[x \mapsto a](y) = \begin{cases} \alpha(y), & y \neq x, \\ a, & y = x \end{cases}$$

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- ▶ $\mathcal{A} \models_{\alpha} (\varphi \rightarrow \psi)$ iff $\mathcal{A} \not\models_{\alpha} \varphi$ or $\mathcal{A} \models_{\alpha} \psi$

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- ▶ $\mathcal{A} \models_{\alpha} (\exists x)\varphi$ iff there is some $a \in u(\mathcal{A})$, $\mathcal{A} \models_{\alpha[x \mapsto a]} \varphi$

Last two cases, α has no effect on the value of x . Thus, assignments matter **only** to free variables.

Examples of Satisfaction

- ▶ $\mathcal{G} = (\{1, 2, 3\}, E^{\mathcal{G}} = \{(1, 2), (2, 1), (2, 3), (3, 2)\})$
 - ▶ For any assignment α , $\mathcal{G} \models_{\alpha} \forall x \forall y (E(x, y) \rightarrow E(y, x))$ iff for every $a, b \in u(\mathcal{A})$, $\mathcal{A} \models_{\alpha[x \mapsto a, y \mapsto b]} (E(x, y) \rightarrow E(y, x))$

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 - ▶ There is an assignment α for which $\mathcal{W} \models_{\alpha} (Q_a(x) \wedge Q_a(y) \wedge S(x, y))$

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- ▶ $\mathcal{W} = abaaa$
 - ▶ There is an assignment α for which $\mathcal{W} \models_{\alpha} (Q_a(x) \wedge Q_a(y) \wedge S(x, y))$
 - ▶ There is no assignment α which satisfies $\exists x \exists y (Q_b(x) \wedge Q_b(y) \wedge x \neq y)$

Examples of Satisfaction

Consider a signature τ containing the binary relation $<$. A totally ordered set satisfies

- ▶ $\forall x \neg(x < x)$
- ▶ $\forall x \forall y \forall z ((x < y \wedge y < z) \rightarrow (x < z))$
- ▶ $\forall x \forall y (x < y \vee y < x \vee x = y)$

The structures $(\mathbb{Z}, <_{\mathbb{Z}})$, $(\mathbb{R}, <_{\mathbb{R}})$ satisfy the above sentences.

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- ▶ Formulae $\varphi(x_1, \dots, x_n)$ and $\psi(x_1, \dots, x_n)$ are **equivalent** denoted $\varphi \equiv \psi$ iff for every $\mathcal{A}$ and α , $\mathcal{A} \models_{\alpha} \varphi$ iff $\mathcal{A} \models_{\alpha} \psi$